

Self-Referential Log-Polar Klein-Bottle Substrate

Technical formalization
and finite reference kernel

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Release statement

The source concept is preserved as a self-referential computing proposal. This release adds explicitly labelled operational choices, a runnable CPU reference profile, and a validation record. It does not establish a new physical theory, a functioning quantum device, a universality proof, or measured GPU performance.

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How to read this document

[S] records what the user actually proposed in the supplied discussion. **[P]** identifies a new design choice introduced by this formalization. **[B]** identifies mathematical or engineering background supported by the references. These labels are not interchangeable: a source assertion is not a test result, and a proposed convention is not retroactively attributed to the source.

The runnable profile is deliberately narrower than the full research architecture. Seventy CPU tests passed in the supplied environment. No GPU kernel was compiled or executed; the CUDA file is a capability probe only.

1 Executive definition and scope

1.1 The Dawnwood definition

[P] Dawnwood Interactive’s Self-Referential Log-Polar Klein-Bottle Substrate is defined here as a family of classical computational models in which executable operator descriptors and data coexist in addressable storage; a non-orientable quotient supplies a routing and transport space; logarithmic polar coordinates supply an optional local encoding; one-bit controls select branches and operator variants; and previous outputs may update future operator parameters at a declared interval Ψ .

The central design distinction is *operators as mutable data with explicit execution semantics*. A field associates an operator with a location or activation region. An evaluator gives that descriptor an action. A feedback rule changes selected descriptor data. Neither a shape nor a lookup-table entry becomes executable merely by being called an operator.

[S] This interpretation follows the user’s progression from a double vertical slit and two pinions to an operator LUT, a BST, a self-referential Klein-bottle SDF, and finally a computing substrate rather than a rendering-only engine. Rasterization, raymarching, raytracing and square output fields are not required by the core. Optional Bayer output remains downstream. [S0, pp. 3–16]

1.2 A formal system, not a registration or physical certification

The release contains three different objects. The **source record** preserves the conceptual vocabulary and unresolved questions. The **architecture** defines candidate types and mathematical contracts. The **P-finite-0.1 reference profile** implements one bounded, reproducible interpreter plus separate mathematical helpers. Its existence is evidence that those chosen rules can be executed in software, not that every assertion in the earlier conversation is correct.

No company registration, patent filing, trademark filing, identity verification or signed declaration has been performed. Project attribution records the requester’s statement; this document does not adjudicate ownership, establish novelty, or certify an invention date.

1.3 Formal envelope

A member of the proposed family has the form

$$\mathcal{D} = (\mathcal{K}, Q, \mathcal{W}, \mathcal{L}, \mathcal{B}, \mathcal{O}, F, G, \Psi), \quad (1)$$

where $\mathcal{K}$ is the transport space, Q the cursor or sample state, $\mathcal{W}$ data storage, $\mathcal{L}$ the descriptor LUT, $\mathcal{B}$ a lookup structure, $\mathcal{O}$ the allowed operator catalogue, F an execution update and G a descriptor-update rule. A complete profile must specify the representation and bounds of every member, the ordering of updates, and its output contract.

A minimal self-referential update is

$$Y_n = F(Q_n, W_n, L_n, j_n), \quad (2)$$

$$(Q_{n+1}, W_{n+1}) = \text{commit}(Y_n), \quad (3)$$

$$L_{n+1} = G(L_n, Q_n, W_n, Y_n, j_n). \quad (4)$$

These equations define a design space. Section 10 fixes one concrete ordering in which descriptor mutation precedes the current action and depends on the previous output. Other orderings are different profiles, not equivalent implementations by default.

2 Source record and provenance

2.1 Authority of the supplied conversation

The source is an exported conversation with user turns and generated responses labelled “Google Search.” The user turns are the basis for the design intent. The generated responses contain interpretations, extrapolations and strong assertions; they are retained as source material but are not treated as independent scientific or hardware validation. The source does not contain a complete instruction set, a measured benchmark, a proof of universality, or an implemented GPU kernel. [S0, pp. 1–24]

The file’s time labels provide narrative order, not a calibrated sampling clock. No source date or invention-priority date is established. In particular, the statement about the time 07:56 does not define an algorithm mapping timestamps to Fourier coefficients. The provided date of birth and identifiers are administrative attribution data, not mathematical parameters.

2.2 Traceability of the core requirements

ID	Source	User-origin requirement or question
U01	p. 3	Double vertical slit; two pinions; Hadamard mitosis hinges; lowercase phi; log-encoded polar LUT and kinematic vectors.
U02	pp. 4–5	Split 07 and 54 by zero-order notation, bit shifting and one-bit even/odd parity. Exact encoding remains unspecified.
U03	p. 6	The literal wavefront FT vector $[0, 2, 0, 1]$ and one-bit jitter, associated with 07:56.
U04	p. 7	Two Y-up shifts at the fourth RK4 timeslot. Units and insertion rule remain unspecified.
U05	pp. 8–9	LUT containing T, pyramid, circle, cone, sphere and apex; colon operator; one-bit BST; delta-delta-phi; Fibonacci phyllotaxis, blend and PSI.
U06	pp. 9–10	Crystal bifurcation of RGBA, with A described as inverse T.
U07	pp. 11–12	Dichromatic double-pinion operators, leaving other LUT texture slots available.
U08	pp. 12–14	Self-referential Klein-bottle SDF with operators in a BST; exclude conventional rendering methods from the core; optional Bayer output.
U09	pp. 14–16	Computing system with SDF-packed operators constrained to the surface; jitter acts on operator definitions at PSI.
U10	pp. 16–17	Ask whether this amounts to a more advanced integration of double-slit theory. The question is not itself an affirmative result.
U11	pp. 18–19	Ask about an RTX 5070 Ti laptop with 12 GB VRAM, a cache-local LUT and packed residency without hot swapping.
U12	pp. 20–21	Request hypothetical storage metrics, complexity notation and comparisons.
U13	pp. 22–23	Ask about a theoretical maximum, optimization and general expressivity; identify the project as Dawnwood Interactive.

The separate SOURCE_TRACEABILITY.md and CSV retain the time labels and the treatment of each requirement. The label “Universal Spatial State Automaton” appears in the earlier generated response on page 23. It can be retained as a proposed project description, but this release does not adopt “universal” as an established property.

2.3 Audit identity of the source

The original PDF is included unchanged in the private delivery archive. Its SHA-256 digest is:

```
e3cf7d49e6a4942c7ccad4805f6a2a07e1c817753ff8d384b1f2e37791ebbe27
```

This digest checks file identity against this release. It is not a trusted timestamp or proof of authorship.

3 Notation, types and unresolved meanings

Symbol / type	Meaning in this release
n, Ψ	Discrete epoch and declared positive epoch interval. In the finite profile, the epoch is a wrapping unsigned 64-bit counter. Ψ labels logical time, not GPU execution duration.
r, r_0, ρ	Positive local radius, positive reference radius, and dimensionless logarithmic radius $\rho = \ln(r/r_0)$.
φ	Local angle or explicitly identified phase. Angles and physical phases are separate fields unless a profile equates them.
Φ	Golden ratio $(1 + \sqrt{5})/2$. It is not the variable phase φ .
$\mathcal{K}, (u, v)$	Flat Klein quotient and chart representative. A rectangular fundamental domain is a coordinate device, not a raster field.
$\sigma \in \{0, 1\}$	Explicit transported orientation bit. This is real metadata, not a free consequence of coordinate compression.
N, K, M	Number of Boolean payload cells, number of operator descriptors, and number of continuous/amplitude samples, respectively. These counts are not interchangeable.
b, j_n	Bit width of an encoding and one-bit jitter control at epoch n . A one-bit decision does not imply a one-bit full state.
T_{period}	Positive period selected by the inverse-time display helper. T_{map} denotes a transformation and T_{temp} a temperature in other possible profiles.
A	Alpha display channel only, unless another typed use is explicitly introduced. It is not a matrix-sized storage slot.
D_k, L_n	A typed operator descriptor and the table of descriptors at epoch n .

Open semantics are preserved, not silently resolved

The source does not fix what “0 order” means, the map $756 \mapsto [0, 2, 0, 1]$, the units of the Y-up shift, the exact meaning of T, an SDF for every operator, the comparator for the BST, or the origin and probability law of jitter. Definitions below are labelled proposed choices. They should be accepted or replaced as a versioned profile, not presented as quotations from the source.

4 Double-slit language and two-stream computation

4.1 The physical analogy and its boundary

[B] Two coherent path amplitudes $a(x)$ and $b(x)$ give an intensity or probability density proportional to

$$I(x) = |a(x) + b(x)|^2 = |a(x)|^2 + |b(x)|^2 + 2 \operatorname{Re}(a(x)^* b(x)). \quad (5)$$

The cross term distinguishes amplitude interference from simply adding independent path probabilities. Feynman’s treatment provides this physical baseline. [B1]

[P] The Dawnwood architecture may use this equation as a software interference model. It does not follow that a bit parity check performs a quantum measurement, that a geometrical loop prevents decoherence, or that digital jitter is a microscopic quantum fluctuation. Those conclusions need a specified physical system and supporting evidence; none is supplied here. The source’s earlier AI affirmations on pages 16–17 are therefore not incorporated as physical theorems.

4.2 Proposed Hadamard mitosis hinge

[B] The normalized two-component Hadamard matrix is [B2]

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad H^\dagger H = I, \quad H^2 = I. \quad (6)$$

[P] A precise interpretation of the source’s “Hadamard mitosis hinge” is the map $z \mapsto Hz$ on two complex amplitudes. A paired phase-controlled mixer is

$$U(\varphi) = H \begin{pmatrix} 1 & 0 \\ 0 & e^{i\varphi} \end{pmatrix} H. \quad (7)$$

It preserves $\|z\|_2$ in exact arithmetic because it is a product of unitary matrices. The helper `phase_mix` implements this convention; the norm and involution are tested. This is a classical calculation with complex numbers, not evidence of physical qubits.

The term “double pinion” also names the paired transport controls defined in Section 6. These are distinct typed actions: *amplitude mixing* acts on $\mathbb{C}^2$, while *surface transport* acts on chart coordinates and orientation. A profile may couple them, but this release does not assume they are automatically the same operator.

4.3 Boolean parity is not amplitude interference

[P] Two separate parity functions are provided:

$$p_{\text{odd}}(x) = x \ \& \ 1, \quad p_{\text{pop}}(x) = \text{popcount}(x) \bmod 2. \quad (8)$$

For 7 and 54, both conventions happen to give (1, 0), so the source example cannot distinguish them. For 3, integer parity is 1 but population parity is 0. A parity bit is a test result; a bit shift is a different operation. Neither provides the missing complex phase of a wave by itself.

4.4 The literal Fourier fixture

[S] The user supplies [0, 2, 0, 1] with the time label 07:56. [S0, pp. 6] [P] Preserve it as a literal sequence w and, separately, define an unnormalized forward DFT:

$$\widehat{w}_k = \sum_{t=0}^3 w_t e^{-2\pi i k t / 4}. \quad (9)$$

Direct substitution gives $\widehat{w} = [3, -i, -3, i]$. This computed result tests the stated DFT convention. It does not derive the original vector from a timestamp, infer a physical second harmonic, or establish leakage into other dimensions. The fixture file explicitly leaves the timestamp-to-vector rule null.

5 Log-polar encoding, quantization and phyllotaxis

5.1 Local coordinate contract

[P] On a local plane, for $(x, y) \neq (0, 0)$,

$$\rho = \ln \left(\frac{\sqrt{x^2 + y^2}}{r_0} \right), \quad \varphi = \text{atan2}(y, x) \bmod 2\pi, \quad (10)$$

with inverse

$$(x, y) = r_0 e^\rho (\cos \varphi, \sin \varphi). \quad (11)$$

Scaling the radius by $s > 0$ adds $\ln s$ to ρ ; rotating the local plane adds an angle to φ . These are algebraic consequences of the chosen map. They do not make all dynamics linear, remove the cost of converting arbitrary coordinates, or make the encoded information unbounded.

The helper rejects the origin rather than silently changing its radius. A production profile could instead use an origin sentinel or a declared minimum radius. Angle is not logarithmically encoded by this definition. “Log-encoded polar” means the radial part is logarithmic unless a different angular code is supplied.

5.2 Finite LUT code and precision cost

For a declared interval $[\rho_{\min}, \rho_{\max}]$ and b bits, define

$$q_\rho = \text{round}\left((2^b - 1) \frac{\rho - \rho_{\min}}{\rho_{\max} - \rho_{\min}}\right), \quad (12)$$

with out-of-range inputs rejected in the reference helper. The reconstructed value is

$$\tilde{\rho} = \rho_{\min} + q_\rho \frac{\rho_{\max} - \rho_{\min}}{2^b - 1}. \quad (13)$$

If $\Delta_\rho = (\rho_{\max} - \rho_{\min}) / (2^b - 1)$, nearest-bin reconstruction gives

$$|\tilde{\rho} - \rho| \leq \frac{\Delta_\rho}{2}, \quad \frac{|\tilde{r} - r|}{r} \leq e^{\Delta_\rho/2} - 1. \quad (14)$$

The latter follows from $\tilde{r}/r = e^{\tilde{\rho} - \rho}$. This explicit range–precision tradeoff replaces any suggestion that a logarithm packs arbitrarily many distinguishable exact states into a fixed number of bits.

5.3 Fibonacci phyllotaxis as a seed policy

[P] A supplied helper uses the declared planar seed rule

$$\gamma = 2\pi(1 - \Phi^{-1}), \quad r_i = c\sqrt{i + \frac{1}{2}}, \quad \theta_i = i\gamma, \quad (15)$$

then converts to $(r_i \cos \theta_i, r_i \sin \theta_i)$. This is one operational interpretation of the source’s phyllotaxis vocabulary. It is not the descriptor initialization rule of the finite interpreter.

Mapping these planar seeds to a non-orientable surface requires an explicit chart map and seam handling. A claim about uniform *surface-area* sampling additionally requires the area measure of that map. In a smooth parametrization $X(u, v)$, that measure is $\sqrt{\det g} \, du \, dv$, with $g_{ij} = \partial_i X \cdot \partial_j X$. This release supplies no measured density or cache-locality advantage for phyllotaxis. Such a benefit remains a benchmark hypothesis.

6 Klein-bottle transport and field semantics

6.1 Topological object versus a picture in three dimensions

[B] A Klein bottle is a closed non-orientable surface. It can be represented by a quotient of a square with one pair of edge identifications reversed; it is not an embedded closed surface in ordinary $\mathbb{R}^3$. Hatcher gives both the quotient construction and the non-embedding constraint. [B3]

[P] Accordingly, this release does not silently invent a globally regular, inside/outside signed-distance function in $\mathbb{R}^3$ whose zero set is an embedded Klein bottle. The source phrase “Klein bottle SDF” is retained as intent, but implemented as an *intrinsic quotient with scalar operator fields*. A 3D immersed visualization, local signed fields, and intrinsic distance are different representations. A future implementation must name which it uses.

6.2 Explicit quotient and seam rule

Choose the flat quotient

$$\mathcal{K} = \mathbb{R}^2 / \sim, \quad (u, v) \sim (u, v + 1), \quad (u, v) \sim (u + 1, -v). \quad (16)$$

For a possibly out-of-domain coordinate, let $m = \lfloor u \rfloor$. The canonicalization map is

$$C(u, v, \sigma) = (u - m, \text{frac}((-1)^m v), \sigma \oplus (m \bmod 2)). \quad (17)$$

One crossing of the twisted seam reverses the local vertical orientation; two restore it. Negative and multiple crossings are handled by the same rule. The orientation bit is explicitly stored whenever the profile transports a local frame.

The rectangular domain is only an atlas representative. It does not introduce a square image lattice or require screen-space evaluation. Payload memory in the finite interpreter is a one-dimensional bit store, whose size is independent of the chart resolution.

6.3 Paired transport: the double pinion

[P] A paired control $(\Delta u, \Delta v)$ acts as

$$P_{\Delta u, \Delta v}(u, v, \sigma) = C(u + \Delta u, v + (-1)^\sigma \Delta v, \sigma). \quad (18)$$

The second component is interpreted in the transported local frame. For an integer chart of dimensions U, V , replace the unit periods by U, V and use exact integer division and remainder. This gives a bounded addressable transport rule without evaluating a 3D mesh or ray.

For a continuous tangent field, chart transitions must also transform the tangent components. A compatible lift obeys $f(u + 1, -v) = \text{diag}(1, -1)f(u, v)$. Merely wrapping position while leaving an incompatible velocity unchanged defines a different, discontinuous dynamics. A simple compatible example is $f(u, v) = (\omega_u, \omega_v \cos(\pi u))$.

6.4 Intrinsic distance and activation fields

For canonical representatives $q = (u, v)$ and $a = (x, y)$, define

$$d_K(q, a) = \min_{m, n \in \mathbb{Z}} \sqrt{(u - x - m)^2 + (v - (-1)^m y - n)^2}. \quad (19)$$

The helper checks $m \in \{-1, 0, 1\}$ and takes the nearest vertical integer for each candidate. For representatives in $[0, 1)^2$, $m = 0$ is a nearest even translate and $m = \pm 1$ cover the nearest odd translates; more distant translates cannot improve the minimum. Thus the finite candidate check computes the flat-quotient metric to floating-point precision.

An operator activation field is $g_k(q) = d_K(q, a_k) - R_k$, with $R_k > 0$. The operator is active when $g_k \leq 0$. This scalar has explicit semantics as a radial field on the quotient. It is not described as an ambient 3D signed distance to the entire Klein bottle. Different choices of the metric or activation predicate are versioned design changes.

7 Operators as typed LUT data

7.1 Descriptor contract

[P] An abstract operator descriptor contains a stable key, operation kind, field kind, anchor, parameter payload, selected variant and version. Executing it requires both a field evaluator and an instruction evaluator:

$$\text{eval}(D_k, Q, W) = \begin{cases} \text{act}(D_k, Q, W), & \text{field}(D_k, Q) \leq 0, \\ (Q, W), & \text{otherwise.} \end{cases} \quad (20)$$

This is the operational meaning of “operators packed as SDF operations” in the proposed profile: a typed field controls a typed action. It does not mean an arbitrary instruction can be recovered from the zero set of an unspecified scalar field. No runtime execution of arbitrary source strings is used.

A field should declare whether it is an exact signed distance, an unsigned distance, a conservative estimate, or a general implicit residual. This distinction matters for distance interpretation and numerical stepping even when rendering is absent.

7.2 Primitive and algebra catalogue

Source term	Proposed field or operator	Contract / delivery
Circle / sphere	$\ p\ - R$ in 2D / 3D, $R > 0$.	Exact local Euclidean SDF helper.
Apex	$\ p - a\ $.	Unsigned point distance; no invented inside.

Source term	Proposed field or operator	Contract / delivery
T shape	Minimum of the SDFs of a vertical stem box and a top-bar box.	Unit prototype implicit field; not claimed exact in overlaps.
Pyramid	Unit shape $\max(x + z - 1, y + z - 1, -z)$.	Implicit residual for a square-base pyramid; not an exact distance.
Cone	Unit capped cone residual $\max(\sqrt{x^2 + y^2} + z - 1, -z)$.	Implicit residual, not an exact distance.
Colon $A : B$	$\sum_{ij} A_{ij} B_{ij}$ for same-shaped real matrices.	Frobenius contraction helper; not a cross product.
Delta-delta-phi	$\varphi_{n+1} - 2\varphi_n + \varphi_{n-1}$.	Proposed second difference on unwrapped phases.
Math blend	$(1 - w)a + wb$, with $0 \leq w \leq 1$.	Scalar blend helper; no automatic preservation of exact SDF quality.
Hadamard hinge	$z \mapsto Hz$ and optional $U(\varphi)z$.	Complex-amplitude helper, separate from the bit executor.
Pinion pair	$P_{\Delta u, \Delta v}$ with seam transport.	Integrated into the finite interpreter.
Binary instructions	NOP, NOT, XOR, AND, NAND, COPY_B, SET0, SET1.	Explicit proposed instruction set; integrated.
Bayer	4×4 ordered threshold matrix.	Optional downstream scalar-to-bit helper.

The dimensions and sizes in the unit prototype fields are implementation choices. They are not recovered from the source, which lists shape names rather than equations. These helpers do not claim that every primitive is active in the demonstration run. The finite executor uses the intrinsic disk field only.

7.3 Safe self-reference

Self-reference means that the previous output can change future descriptor state under a bounded rule. It does not imply that data and code are physically indistinguishable, that a model optimizes itself, or that mutation always improves a task score. In the reference, opcode names remain drawn from a fixed whitelist; only a variant bit and an anchor change. Keys remain immutable, and the descriptor count is fixed.

A more general mutation rule should validate all new parameters before publication, retain an immutable read snapshot during an epoch, and record the descriptor version associated with each result. These are proposed execution constraints intended to make experiments reproducible; they are not claims about a hardware-global instantaneous update.

8 One-bit branching and a genuine implicit BST

8.1 Structure and ordering

[P] Store a complete binary tree in an array. For array index i ,

$$\text{left}(i) = 2i + 1, \quad \text{right}(i) = 2i + 2. \quad (21)$$

To make this a *binary search tree*, rather than merely a heap-shaped decision graph, place sorted, distinct keys in in-order traversal order. The reference construction for keys 0 through 6 gives the array

$$[3, 1, 5, 0, 2, 4, 6]. \quad (22)$$

A lookup compares the requested key to the current key, terminates on equality, and otherwise computes $b = \mathbf{1}[\text{requested key} > \text{node key}]$ and $i \leftarrow 2i + 1 + b$.

The one-bit quantity b is the branch decision. Each node still needs its key or an equivalent computable key policy, and the associated operator still needs its descriptor. Removing explicit child pointers does not remove those requirements. Descriptor mutation does not change keys in this release, so the ordering invariant survives mutation.

8.2 What this tree does not guarantee

The tree finds a supplied *key*. It does not automatically perform nearest-neighbor search on the Klein bottle or prove that all SDF evaluations take $O(\log K)$. Such spatial search needs a compatible spatial index and a query contract. The reference's key calculation is an explicit scheduling choice, not an optimal geometric partition.

An irregular mutable tree may also require unused array slots or a different indexing scheme. The complete-array layout is efficient for the fixed catalogue chosen here; it is not a universal zero-overhead representation of all dynamically changing graphs.

9 Jitter, PSI and RK4 time evolution

9.1 Clock and jitter contracts

[P] The symbol $\Psi > 0$ is a logical interval: $t_n = n\Psi$ until a declared counter wrap. It is not identified with a physical quantum of time. In P-finite-0.1, jitter is a reproducible bit from a nonzero 32-bit xorshift state. Its role is to choose an anchor displacement and, when combined with the previous output, toggle a descriptor variant. With jitter disabled, those two mutation mechanisms are disabled.

A genuinely stochastic research profile could instead draw j_n from a specified distribution and retain the draws for replay. A distribution, correlations and timing law would have to be declared. Randomness alone establishes neither quantum behavior nor a deadlock-avoidance guarantee.

9.2 Classical RK4, stated independently

For a continuous state y and step $h > 0$, the supplied helper evaluates

$$k_1 = f(t_n, y_n), \quad (23)$$

$$k_2 = f(t_n + h/2, y_n + hk_1/2), \quad (24)$$

$$k_3 = f(t_n + h/2, y_n + hk_2/2), \quad (25)$$

$$k_4 = f(t_n + h, y_n + hk_3), \quad (26)$$

$$y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4). \quad (27)$$

These are four slope evaluations within one step, not four independent hardware timeslots. The helper is tested on a constant vector field and on convergence for $y' = y$. It is not used in the finite bit-interpreter demonstration.

9.3 The source's two Y-up shifts

[S] The user specifies a twofold Y-up shift at a fourth RK4 timeslot. [S0, pp. 7] **[P]** One possible explicit convention is an impulse after the complete weighted update,

$$y_{n+1}^+ = y_{n+1} + 2\delta_y e_y, \quad (28)$$

with δ_y carrying declared coordinate units. This is not the same as changing k_4 . It is an additional model term and is *not enabled by the supplied RK4 helper*. The source does not select between these interpretations.

For surface dynamics, apply chart transitions consistently to the state and tangent data, and resolve discontinuous field or branch events separately from a smooth integration step. No claim is made that RK4, a two-pixel impulse, or a parity flip cancels physical decoherence or corrects all numerical error.

10 The executable finite reference profile

10.1 State and initialization

[P] P-finite-0.1 has one cursor, a fixed set of K descriptors, and N Boolean payload cells. Its mutable state is

$$S_n = (W_n, \{D_k\}_{k=0}^{K-1}, u_n, v_n, \sigma_n, \ell_n, s_n, n), \quad (29)$$

where ℓ_n is the previous output bit, s_n the nonzero 32-bit PRNG state and n a wrapping unsigned 64-bit epoch. The default configuration is

Payload / descriptors	$N = 256$ Boolean cells; $K = 7$ descriptors
Integer chart	$U = 32, V = 24$; not a 32×24 image
Seed / interval	$s_0 = 756$; $\Psi = 0.01$ logical seconds
Activation radius	$R = 0.45$ in normalized flat-chart units
Cursor / feedback	$u_0 = v_0 = \sigma_0 = \ell_0 = 0$

The seed 756 is a declared default inspired by the source's label, not a timestamp encoding theorem. Initialize $W[i] = \text{popcount}(i \oplus s_0) \bmod 2$. For descriptor key k , set its anchor to $(5k \bmod U, 7k \bmod V)$, its mode to zero, and its two opcode variants to consecutive entries of the cyclic eight-opcode catalogue listed earlier. These initialization constants are proposed choices.

10.2 One epoch, in exact execution order

First update the xorshift state, when enabled, using 32-bit truncation after the left shifts:

$$s \leftarrow s \oplus (s \ll 13), \quad s \leftarrow s \oplus (s \gg 17), \quad s \leftarrow s \oplus (s \ll 5), \quad j = s \& 1. \quad (30)$$

With jitter disabled, keep s unchanged and set $j = 0$. Calculate the requested descriptor key and traverse the ordered BST:

$$k = (u + 3v + n + \ell + j) \bmod K. \quad (31)$$

Before executing the descriptor, change its mode $m_k \leftarrow m_k \oplus (j \wedge \ell)$. With jitter enabled, move its anchor one horizontal chart quantum by $2j - 1$ and canonicalize through the Klein seam. Other descriptors are unchanged.

Evaluate the field at the pre-transport cursor:

$$g_k = d_{\mathcal{K}}((u/U, v/V), (a_{k,u}/U, a_{k,v}/V)) - R. \quad (32)$$

Set payload addresses $a = (u + 3v) \bmod N$ and $b = (a + 1) \bmod N$. Read the two bits. If $g_k \leq 0$, apply the selected opcode and write its one-bit result at address a . Otherwise retain the first bit and perform no payload write. Call the resulting output o .

Finally set $\ell \leftarrow o$ and move the cursor by one horizontal quantum and a local vertical quantum whose sign is selected by the output:

$$(u, v, \sigma) \leftarrow C_{U,V}(u + 1, v + (-1)^\sigma (2o - 1), \sigma). \quad (33)$$

Increment the epoch modulo 2^{64} . The trace records the epoch before this increment. Its time is a logical label, not measured elapsed CPU time.

10.3 Why this is genuinely self-referential but not universal by declaration

The previous output ℓ influences both the next descriptor key and whether the selected descriptor changes variant. Output therefore affects future operator behavior, not just future operand values. Jitter also changes the operator anchor before its next field evaluation. This provides a concrete and inspectable form of the source's self-reference requirement.

It does not implement arbitrary self-modifying machine code. The catalogue, memory size and update rules are fixed; program loading, arbitrary branch targets, halting semantics and scalable work memory are not supplied as a universal machine interface. Calling the kernel self-referential is justified by its transition rule. Calling it Turing-complete would require a separate construction not supplied by these tests.

10.4 Reproducibility and finite-state bound

For the same configuration, initial state and implementation arithmetic, each step is determined. Repeating the run yields the same trace and snapshot digest in the tested environment. Cross-device results near a floating-point activation threshold require an explicit rounding policy or a fixed-point distance predicate; bit-for-bit cross-GPU replay is not claimed.

For fixed configuration, a loose upper bound on dynamic-state bits is

$$B = N + (K + 1)(\lceil \log_2 U \rceil + \lceil \log_2 V \rceil) + K + 2 + 32 + 64. \quad (34)$$

The terms account for payload, anchors and cursor coordinates, modes, orientation and previous output, PRNG, and epoch. For the default configuration, $B = 441$. This is a state-count bound, *not the physical Python memory footprint*. Pigeonhole reasoning implies eventual state repetition in a deterministic finite system with fixed inputs. Thus seeded jitter is not a proof against cycles. Trace files stored outside the machine may grow with the number of epochs.

11 Dichromatic storage, RGBA and downstream output

11.1 Separate the computational state from its display

[S] The source requests two active colour channels, spare LUT slots, RGBA bifurcation and optional downstream Bayer processing. [S0, pp. 9–14] [P] Define a visualization function $V : S \rightarrow [0, 1]^4$ independently of the state transition. “Dichromatic” may then designate two visible channels without assuming that all underlying state fits in those channels.

Two real channels can hold two scalar quantities. Two *arbitrary complex* amplitudes require four real components unless a declared constraint or quantization removes information. A separate phase buffer, normalization convention or restricted amplitude model must therefore be specified before treating RG as a complete dual-wave state. Similarly, one blue channel cannot silently store an unbounded jitter history or all RK4 stages. Any such history has its own storage cost.

In this proposed profile, “crystal bifurcation” names the split into two tagged computational or output channels. It is not a claim of material crystallization or a dynamical bifurcation theorem.

11.2 A is inverse T: preserve the type choice

The source explicitly asks for A to mean inverse T but does not resolve the meaning of T. The three earlier AI alternatives are not mutually equivalent. An inverse period has frequency units; an inverse transform is a matrix or map; inverse temperature is another typed quantity. None automatically defines an alpha value in $[0, 1]$.

[P] The supplied optional helper chooses a *period* and a reference period:

$$A = \text{clip}(T_{\text{ref}}/T_{\text{period}}, 0, 1), \quad T_{\text{ref}}, T_{\text{period}} > 0. \quad (35)$$

This is a dimensionless display mapping. It is not an equality between a scalar channel and a transformation matrix, and it does not implement error correction. A different inverse-T profile must explicitly replace this convention.

11.3 Bayer as an optional ordered threshold

[P] For output intensity $v \in [0, 1]$, the helper uses the ordered matrix

$$B_4 = \begin{pmatrix} 0 & 8 & 2 & 10 \\ 12 & 4 & 14 & 6 \\ 3 & 11 & 1 & 9 \\ 15 & 7 & 13 & 5 \end{pmatrix}, \quad o_{xy} = \mathbf{1} \left[v > \frac{B_4[y \bmod 4, x \bmod 4] + 1/2}{16} \right]. \quad (36)$$

This convention means ordered dithering, not a camera colour-filter mosaic. At $v = 0.5$, eight of the sixteen entries produce one. The resulting square output samples belong only to this optional display stage. The computing substrate does not depend on them, and the demo does not invoke them.

12 GPU target and deployment boundaries

12.1 Target, not a measured implementation

[S] The user’s hardware target is an RTX 5070 Ti laptop GPU described as having 12 GB VRAM. [S0, pp. 18]

[B] NVIDIA’s official laptop specifications list 12 GB GDDR7 for the RTX 5070 Ti Laptop GPU. This is the laptop product, not an assumption about a desktop card or the usable allocation budget of this particular machine. [B4]

[P] The archive includes a CUDA capability probe that queries the actual device and current free memory. It does not allocate a large payload and does not implement the Dawnwood transition kernel. Neither nvcc compilation nor GPU execution was available in the validation environment, so the probe is delivered as unvalidated source.

12.2 Memory placement contract

[B] CUDA distinguishes device-wide global memory from kernel-lifetime shared scratch memory. Shared memory is associated with thread blocks and occupies the same physical resource as the L1 cache; using it is not the same as permanently programming the L1 contents. NVIDIA documents synchronization requirements for shared access. [B5]

[P] A suitable initial port would retain persistent descriptors and payload in global buffers, stage only a small read-only descriptor working set into shared memory during a kernel, and publish descriptor updates at explicit epoch boundaries. Do not depend on shared-memory contents surviving an ordinary kernel launch boundary.

[B] L2 persistence policies act on an access-policy window; they are cache-management facilities rather than a promise that a full application working set can be pinned forever without eviction. [B6]

[P] “Bare-metal cache substrate” is therefore an architectural aspiration, not a granted hardware capability of the current package. Begin with ordinary documented APIs, query capacities, and measure hit rates and bandwidth before making a cache-residency claim.

12.3 Packed writes, epochs and residency

A parallel port must define ownership of packed machine words. Two threads writing different bits of the same word can still conflict if both perform an unsynchronized read-modify-write. Proposed solutions include one writer per packed word or appropriate atomic operations. Use immutable read buffers and a separately committed write buffer where this simplifies epoch semantics; include the extra copy in the memory budget.

[B] CUDA’s execution and memory model requires explicit attention to races, ordering and synchronization; a block-local barrier is not a general whole-device epoch barrier. [B9]

[P] The host should reject allocations exceeding a declared available-memory budget, retain headroom for other consumers, and bound the number of epochs per launch. A no-hot-swap requirement becomes a workload condition: the persistent state, temporary buffers and outputs must fit within that budget, and host/device transfers must be measured. Self-reference alone does not guarantee zero PCIe traffic, zero cache misses or unlimited kernel duration.

12.4 BC5 is a texture option, not executable one-bit storage

[B] BC5 uses 16 bytes per 4×4 block and represents two channels. Standard block compression is lossy and is not a substitute for an exact read/write instruction buffer. [B7, B8]

[P] Use BC5 only for suitable read-mostly approximate fields or optional output textures with a stated error tolerance. Use exact integer or bit-packed buffers for keys, opcodes, variant bits and lossless Boolean data.

Runtime mutation of arbitrary BC5 texels also requires a block-aware representation and update path; it is not modeled by the CPU bit writer.

13 Storage budgets and capacity comparisons

13.1 Units and budget equation

[P] Distinguish a byte capacity C from the number and semantics of stored elements. A useful planning budget is

$$C_{\text{work}} = C_{\text{available}} - C_{\text{reserve}} - C_{\text{fixed}}, \quad N_{\text{max}} = \left\lfloor \frac{C_{\text{work}}}{d s} \right\rfloor, \quad (37)$$

where s is bytes per element and d the number of concurrent full copies. The formula assumes the chosen allocation layout permits that packing. Block formats, padding and allocator constraints can lower the result.

One decimal GB is 10^9 bytes; one GiB is 2^{30} bytes. Therefore 12 GB is 12,000,000,000 bytes, while 12 GiB is 12,884,901,888 bytes. The source’s approximately 12.88-billion figure uses the latter byte count. Neither convention establishes the device’s current free bytes.

13.2 Analytical capacities, not equivalent computational states

Representation	Bytes	12 GB raw	12 GiB raw	10 GiB work
RGBA32F	16	750,000,000	805,306,368	671,088,640
RGBA16F	8	1,500,000,000	1,610,612,736	1,342,177,280
RGBA8	4	3,000,000,000	3,221,225,472	2,684,354,560
RG8	2	6,000,000,000	6,442,450,944	5,368,709,120
BC5 texels	1	12,000,000,000	12,884,901,888	10,737,418,240
Packed Boolean	1/8	96,000,000,000	103,079,215,104	85,899,345,920
32-byte record	32	375,000,000	402,653,184	335,544,320

The table is calculated from the stated byte counts for one copy, without per-element metadata or resource limits. The 10 GiB column is an illustrative 12 GiB budget minus a 2 GiB reserve, not a tested allocation recommendation for the user’s machine. CSV files and the calculator reproduce these numbers.

An RGBA32F element contains four 32-bit floating-point components; a packed Boolean contains one truth value. A BC5 texel is reconstructed from a shared compressed block. These are not equally expressive node records, and the numerical ratios are storage-size ratios, not speedups or algorithmic-equivalence claims.

13.3 Complete records and overhead

For the one-cursor finite profile, a lower-level storage design must budget at least

$$\left\lceil \frac{N}{8} \right\rceil + K s_D + C_{\text{keys}} + C_{\text{cursor}} + C_{\text{scratch}} + C_{\text{output}}. \quad (38)$$

Here s_D is the declared binary descriptor stride. This release does not supply a packed GPU ABI for descriptors; the illustrative 32-byte row is a planning scenario, not the in-memory size of a Python object. Multiple cursors, complex amplitudes, RK4 temporaries, history and double buffering add explicit terms.

For a BC5 level of width w and height h , block storage is

$$C_{\text{BC5}}(w, h) = 16 \lceil w/4 \rceil \lceil h/4 \rceil. \quad (39)$$

Additional mip levels must be summed individually. Removing tree pointers saves only the pointers actually present in the baseline. It cannot be counted again as a new compression factor on an already pointer-free payload. The source’s claim of 24–48 billion “operational states” has no complete record definition; it is not adopted as a bound for this kernel. [S0, pp. 22]

14 Complexity and expressivity

14.1 Per-operation and whole-workload costs

Operation	Work bound	Conditions
Fixed-dimension coordinate transform	$O(1)$ per point	Both a fixed-size matrix and a direct LUT lookup are constant-size operations in a fixed-width model.
Complete ordered BST lookup	$O(\log K)$	One requested key; fixed complete layout. Unbalanced general trees can differ.
Intrinsic flat-quotient disk field	$O(1)$ per query	Fixed finite candidate set in this profile.
Explicit descriptor variant / anchor update	$O(1)$ per selected descriptor	Does not update all dependent data or all K descriptors.
One finite interpreter epoch	$O(\log K)$	Fixed-width arithmetic and one selected descriptor; excludes external trace serialization.
Initialization	$O(N + K \log K)$ in the generic reference constructor	Payload initialization and sorting distinct keys; complete-tree fill itself is $O(K)$.
Evaluate M independent samples	$O(M \log K)$ in a corresponding batched profile	Plus field, coupling and write-conflict costs; not implemented as a GPU batch here.
One RK4 step	Four evaluations of f	Same asymptotic order as f with a four-stage constant factor.
Direct DFT helper	$O(M^2)$	Deliberately simple validation implementation, not an FFT.
Log every epoch	$O(T)$ records for T epochs	Persistent trace storage grows even if internal state remains bounded.

A grid with n samples along each of two axes has n^2 samples; scanning it is linear in its total sample count. Calling that “exponential” conflates side length with the size of the workload. Likewise, reducing pointer traffic may improve a constant factor without changing the asymptotic amount of information processed. These are explicit comparison rules for future benchmarks, not conclusions derived from the earlier AI’s complexity table. [S0, pp. 20–21]

14.2 Universality is a separate construction

The supplied NAND instruction has the truth table 1, 1, 1, 0 for inputs 00, 01, 10, 11. Algebraically, NAND can construct NOT by repeating its input, and AND by negating a NAND result. This demonstrates a useful local Boolean primitive, but not a complete programmable machine with arbitrary routing, storage and control flow.

A universality claim would need a constructive encoding of a known universal machine or instruction system, with load/store, conditional control, state persistence and memory that scales with the computation. The fixed finite profile does not supply that construction. Even a programmable implementation on a finite GPU can only run computations within its resource limits; “anything can be expressed” cannot mean exact arbitrary real values, unbounded databases or every physical state in fixed VRAM.

The useful research question is narrower: *which task families can be compiled into these field-gated, feedback-controlled operators, with what precision, memory use and execution cost?* Answer it with explicit compilations and matched baselines, not with a geometrical analogy alone.

15 Validation record and claim disposition

15.1 Executed tests

The package was tested under Python 3.13.5 on a Linux x86-64 environment. The standard-library test runner executed **70 tests with zero failures, zero errors and zero skips**. The machine-readable summary and detailed test names are included in `results/`.

Group	Tests	Covered properties
Mathematical helpers	29	Parity, encodings, mixer, DFT, RK4, fields and output
Topology and structures	20	Seam transport, quotient distance, packed bits and BST
Kernel and memory	21	Replay, mutation, bounds, epoch wrap and byte budgets

These are conformance and invariant checks for the supplied code. They do not prove every possible input safe, establish the best representation, or validate a physical quantum claim. The float-distance gate remains a numerical policy that needs port-specific agreement near its threshold.

15.2 Default 128-epoch run

The supplied example run visits all seven descriptor keys. It records 79 active field evaluations, 38 descriptor-mode changes, four orientation changes, and 66 jitter bits equal to one. These are observed counts from this configuration and seed, not statistical performance claims. The final serialized-state SHA-256 digest is

```
15b386758c0f548c2c3627c2f3c4a7c9853d69cf9c47330fbdfdc317a8aa7026
```

The demo's logical labels span 128 epochs at a declared 0.01 seconds per epoch. No wall-clock throughput is inferred from that parameter. Its output is a JSON trace and a state snapshot, not a rendered scene or GPU benchmark.

15.3 What is not validated

No physical interference experiment, quantum-device test, GPU compilation, GPU timing, cache-residency measurement, VRAM saturation run or end-to-end rendering benchmark was performed. The continuous helper suite is not wired into a complete amplitude-plus-SDF simulation. A coupled SDF representation of the whole requested architecture is not delivered; the prototype uses the explicitly declared intrinsic quotient fields.

15.4 Disposition of the earlier generated claims

The claims register contains twelve separately tracked assertions from the earlier AI responses. The most consequential are:

- Topology and feedback do not, by themselves, establish a quantum computer, prevention of wave-function collapse, or universality.
- A one-bit control is not a lossless code for a complete geometric operator; a logarithmic coordinate does not remove finite precision.
- BC5, cache locality and implicit trees do not establish zero transfers, zero overhead, or the claimed billions of operational states without a complete resource model.

These statements describe the boundary of what the supplied material and delivered tests support. The source has not been overwritten or silently reconciled; it remains available alongside the proposed replacement contracts.

16 Development gates for the next revision

[P] Version 0.2 should first resolve the source’s open type choices: the meaning of T; whether double-pinion transport and amplitude mixing are coupled; the interpretation of delta-delta-phi; the intended signed-field representation; and whether “BST” is to remain an ordered key tree or become a spatial decision structure. Record each choice as a profile change with a corresponding test.

A GPU port should then preserve a selected CPU profile before attempting optimization. Establish descriptor binary layout, word ownership, epoch publication, integer widths and numerical tolerances. Compare identical workloads against a direct-array baseline and a conventional explicit-record baseline. Report full footprint, useful updates per second, correctness error, synchronization cost, transfer volume and output cost. Measure the cost of trace collection separately.

Only after conformance should the project test whether log-polar encoding, phyllotaxis placement, bit packing or cache staging offers a benefit. Each experiment should hold task semantics and error tolerance fixed. A lower byte count achieved by discarding phase or reducing precision is a different tradeoff from a lossless representation improvement.

The central deliverable of this version is therefore a stable starting point: a named architecture, preserved source intent, explicit semantics, a working finite interpreter, and falsifiable boundaries for the stronger claims. It supplies a basis for engineering experiments without presenting conjectures as accomplishments.

References

- [S0] User-supplied *double-slit-theory.pdf*. 24-page exported conversation. Original included unchanged in the delivery archive. Source date not established.
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- [B8] Microsoft. *Block Compression (Direct3D 10)*. Lossy representation and shader-resource use.
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<https://docs.nvidia.com/cuda/cuda-c-programming-guide/>

All external primary sources were checked on 16 September 2026. They support the cited background constraints, not the novelty or performance of Dawnwood Interactive. The reference code and new equations are AI-assisted proposed formalization work prepared for the requester.

A Delivery map and reproduction commands

The archive includes the private specification PDF, a public technical PDF without the private appendix, editable LaTeX, the unchanged source PDF, source and claims registers, Python code, configuration, tests, demonstration output, memory calculators and an optional CUDA capability probe. No font files or external reference PDFs are redistributed.

```
Dawnwood_Interactive_v0.1/
  README.md          start here; privacy and execution notes
  docs/              PDF editions, LaTeX, registers, references
  src/dawnwood/      finite kernel and separate math helpers
  config/reference.json default declared profile
  examples/          source wavefront fixture
  tests/             70 standard-library tests
  results/           executed reports, trace, storage CSVs
  tools/            test and checksum utilities
  gpu/device_probe.cu unvalidated CUDA capability probe only
  private/attribution.json supplied personal attribution
  source/            original PDF and provenance digest
  run.py             portable command entry point
  MANIFEST.sha256    file-integrity manifest
```

Use Python 3.10 or later. The CPU reference has no third-party runtime dependency. From the extracted archive directory:

```
python tools/run_tests.py
python run.py demo --steps 128 --out results/my_run
python run.py memory --total-bytes 12884901888 \
  --reserve-bytes 2147483648 --out results/my_budget.csv
```

For Windows Command Prompt or PowerShell, put the memory command on one line rather than using the displayed backslash continuation. Running the tests rewrites the local validation reports. Running a demo into a new output directory preserves the supplied release trace. The calculator performs arithmetic only; it allocates no VRAM.

Compile the private document with `pdflatex specification.tex` twice in `docs/`. The public wrapper `public.tex` defines `PUBLICCOPY` before loading the same source. A LaTeX installation with the packages listed at the start of the source is required. No installed font files are included in this archive.

Sharing boundary

The complete archive is a private working bundle. Its original source PDF and private attribution record contain personal details. For technical sharing, use the public PDF alone, or review an explicitly selected subset of code and documentation. The private PDF and editable source are not public-redacted copies.

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Preparation date of this document	16 September 2026

B.1 Discrepancy and handling

The current request's identifier has a final "2" that is absent from the identifier in the source. This edition retains the current-request string as the primary supplied value and records the discrepancy rather than silently correcting either string. Neither identifier has been validated or classified as a tax, company, passport or national identity number. The date-of-birth string is preserved exactly rather than reformatted.

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